

CHAPTER 1

HVAC SYSTEM ANALYSIS AND SELECTION

<i>Selecting a System</i>	1.1	<i>Security and Environmental Health and Safety</i>	1.9
<i>HVAC Systems and Equipment</i>	1.4	<i>Automatic Controls and Building</i>	
<i>Space Requirements</i>	1.6	<i>Management Systems</i>	1.9
<i>Air Distribution</i>	1.8	<i>Maintenance Management</i>	1.10
<i>Pipe Distribution</i>	1.9	<i>Building System Commissioning</i>	1.10

AN HVAC system maintains desired environmental conditions in a space. In almost every application, many options are available to the design engineer to satisfy a client’s building program and design intent. In the analysis, selection, and implementation of these options, the design engineer should consider the criteria defined here, as well as project-specific parameters to achieve the functional requirements associated with the project design intent. In addition to the design, equipment, and system aspects of the proposed design, the design engineer should consider sustainability as it pertains to responsible energy and environmental design, as well as constructability of the design.

HVAC systems are categorized by the method used to produce, deliver, and control heating, ventilating, and air conditioning in the conditioned area. This chapter addresses procedures for selecting an appropriate system for a given application while taking into account pertinent issues associated with designing, building, commissioning, operating, and maintaining the system. It also addresses specific owner requirements and constraints associated with selecting the optimum HVAC system for the application. Chapters 2 to 18 describe specific approaches and systems along with their attributes, based on their heating and cooling medium, the commonly used variations, constructability, commissioning, operation, and maintenance.

This chapter is intended as a guide for the design engineer, builder, facility manager, and student needing to know or reference the analysis and selection process that ultimately leads to recommending the optimum system for the job. The approach applies to HVAC equipment conversions, building system upgrades, system retrofits, building renovations and expansion, and new construction for any building: small, medium, large, below grade, at grade, low-rise, and high-rise. This system analysis and selection process (Figure 1) approach helps guide the design engineer in drafting a report that recommends the best system(s) for any building program, regardless of facility type. This chapter’s analysis examines objective, subjective, short-term, and long-term goals along with constraints. Figure 1 also highlights five project delivery methods: performance contracting, design-bid-build, design-build, integrated building design, and construction management.

1. SELECTING A SYSTEM

The design engineer is responsible for considering various systems and equipment and recommending one or more system options that will meet the project goals and perform per the design intent. It is imperative that the design engineer and owner collaborate to identify and prioritize criteria associated with the design goals. In addition, if the project includes preconstruction services, the designer, owner, and operator should consult with a builder to take advantage of a constructability analysis as well as the consideration of value-

engineered options. Occupant comfort (as defined by ASHRAE Standard 55), process heating, space heating, cooling, and ventilation criteria must be considered when selecting the optimum system(s), as well as the following:

- Temperature
- Humidity
- Air motion
- Air/water velocity
- Water quality and/or reuse
- Outdoor air quality or purity
- Indoor air purity or quality
- Air changes per hour
- Acoustics and vibration
- Local climate
- Mold and mildew prevention
- Capacities (existing, proposed, and future expansion)
- Redundancy
- Spatial requirements (present and future)
- Environmental health and safety design
- Security
- First cost
- Return-on-investment cost
- Energy consumption costs
- Operator labor costs
- Maintenance costs
- Serviceability
- Reliability
- Flexibility
- Controllability
- Replaceability
- Life-cycle analysis
- Sustainability of design
- Seismic protection
- Filtration and filtration effects
- Changing codes and standards

Because these factors are interrelated, the owner, design engineer, operator, and builder must consider how these criteria affect each other. The relative importance of factors such as these varies with different owners, and can often change from one project to another for the same owner. For example, typical owner concerns include first cost compared to operating cost, extent and frequency of maintenance and whether that maintenance requires entering the occupied space, expected frequency of system failure, effect of failure, and time required to correct the failure. Each concern has a different priority, depending on the owner’s goals.

Additional Goals

In addition to the primary goal of providing the desired environment, the design engineer should be aware of and account for other goals the owner may require. These goals may include the following:

The preparation of this chapter is assigned to TC 9.1, Large Building Air-Conditioning Systems.

- Seasonal start-up date
- Occupant move-in date
- Operator training
- Supporting a process, such as operation of computer equipment
- Promoting a germ-free environment
- Increasing marketability of rental spaces
- Increasing net rental income
- Increasing property saleability
- Public image of the property
- Certification of energy use (LEED®, ENERGYSTAR®, etc.)

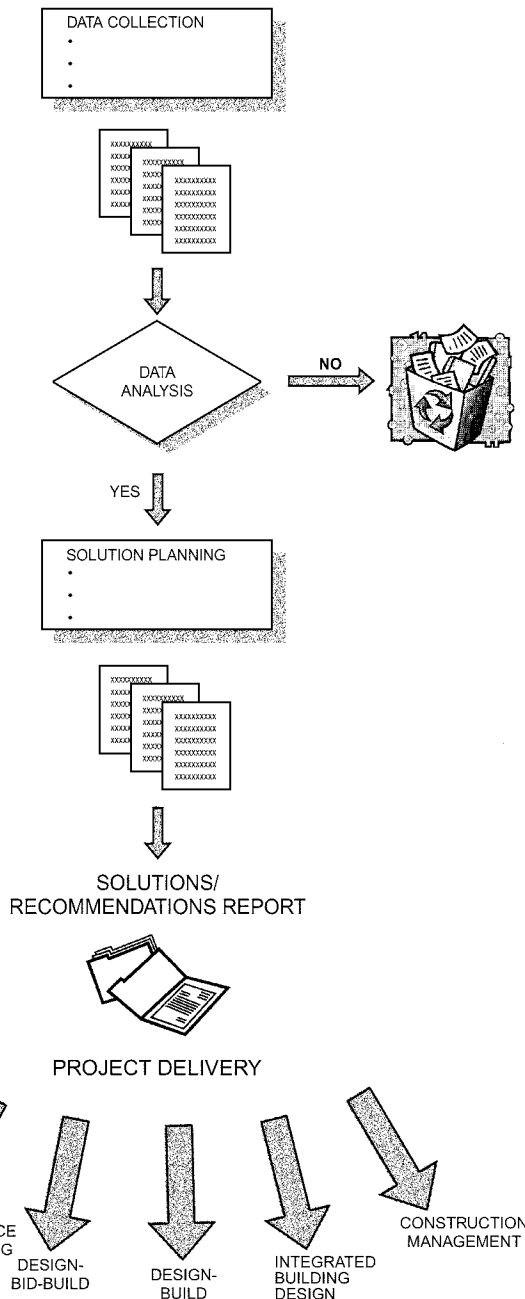


Fig. 1 Process Flow Diagram
(Courtesy RDK Engineers)

The owner can only make appropriate value judgments if the design engineer provides complete information on the advantages and disadvantages of each system option. Just as the owner does not usually know the relative advantages and disadvantages of different HVAC systems, the design engineer rarely knows all the owner's financial and functional goals. Hence, the owner must be proactive and involved in system selection in the conceptual phase of the job. The same can be said for operator participation so that the final design intent is sustainable and can continue to fulfill design intent.

All owners should request and/or require the design team to provide building and HVAC security (as defined in Chapter 61 of the 2019 *ASHRAE Handbook—HVAC Applications*). This security, in addition to environmental health and safety criteria, should address outside-the-building influences (e.g., smoke, toxic fumes, flood, etc.) as well as influences inside the building. System selection pertaining to HVAC security may require another design team member, a security consultant, and/or the owner's own security group.

Equipment and System Constraints

Once the goal criteria and additional goal options are verified, equipment and system constraints must be assessed and documented as part of the system analysis and selection recommendation report. These constraints may include the following:

- Performance limitations (e.g., temperature, humidity, space pressure, etc.)
- Code updates and/or new codes
- Available capacity including equipment size, as well as ductwork and/or pipe size
- Available space and access in and/or out with new or old equipment
- Available utility source
- Energy budget (e.g., conceptual Btu/h-ft² per year), code-driven or targeted
- Equipment efficiency
- Operator knowledge and capabilities
- Existing building occupants
- Building architecture

Few projects allow detailed quantitative evaluation of all alternatives. Common sense, historical data, and subjective experience can be used to narrow choices to one or two potential systems. Heating and air-conditioning loads often contribute to constraints, narrowing the choice to systems that fit in available space and are compatible with building architecture. Chapters 17 and 18 of the 2017 *ASHRAE Handbook—Fundamentals* describe methods to determine the size and characteristics of heating and air-conditioning loads. By establishing the capacity requirement, equipment size can be determined, and the choice may be narrowed to those systems that work well on projects within the required size range.

Loads vary over time based on occupied and unoccupied periods, changes in weather, type of occupancy, activities, internal loads, and solar exposure. Each space with a different use and/or exposure may require its own control zone to maintain space comfort. Some areas with special requirements (e.g., dehumidification requirements) may need individual systems. The extent of zoning, degree of control required in each zone, and space required for individual zones also narrow system choices.

No matter how efficiently a particular system operates or how economical it is to install, it can only be considered if it (1) maintains the desired building space environment within an acceptable tolerance under expected conditions and occupant activities and (2) physically fits into, on, or adjacent to the building without causing objectionable occupancy conditions.

Cooling, heating, and humidity control are often the basis of sizing HVAC components and subsystems, but ventilation requirements can also significantly impact system sizing. For example, if

large quantities of outdoor air are required for ventilation or to replace air exhausted from the building, the design engineer may only need to consider systems that transport and effectively condition those large outdoor air volumes, thus eliminating many of the other HVAC systems (e.g., unit ventilators, through-the-wall window units) from system consideration. The cooling, heating, humidification, and dehumidification delivery performance compromises may be minor for one application in a given climate, but may be unacceptable in another that has more stringent requirements.

HVAC systems and associated distribution systems often occupy a significant amount of space. Major components may also require special support from the structure. The size and appearance of terminal devices (e.g., grilles, registers, diffusers, fan-coil units, radiant panels, chilled beams) affect architectural design because they are visible in the occupied space, which constrains the design.

Sustainable energy consumption can be compromised and long-term project success lost if building operators are not trained to efficiently and effectively operate and maintain the building systems. For projects in which the design engineer used some form of energy software simulation, the resultant data should be passed on to the building owner so that goals and expectations can be measured and benchmarked against actual system performance. Even though the HVAC designer's work may be complete after system commissioning and turnover to the owner, continuous acceptable performance is expected and will only occur if the owner understands the background and systems selected and their anticipated performance. See ASHRAE *Guideline 0* and to ASHRAE's Building Energy Quotient (bEQ) program (www.buildingeq.com).

System operability should be considered in system selection. Constructing a highly sophisticated, complex HVAC system in a building where maintenance personnel lack the required skills for maintenance can be a recipe for disaster at worst, and at best may require the use of costly outside maintenance and service contractors to achieve successful system operation.

The design engineer should closely coordinate the system constraints with the rest of the design team, as well as the owner, to overcome design obstacles associated with the HVAC systems under consideration for the project.

Constructability Constraints

The design engineer must take into account HVAC system constructability issues before the project reaches the construction document phase. Some of these constraints may significantly affect the success of the design and cannot be overlooked in the conceptual and design development phases. Some issues and concerns associated with constructability are

- Existing conditions (e.g., floor load, access into and through a building)
- Rigging equipment into and out of a building to a designated area
- Demolition and impact on adjacent space and existing systems in operation
- Maintaining existing building occupancy, use of the building, and system operation
- Ability to phase HVAC system installation
- Temporary HVAC
- Equipment availability (e.g., delivery lead time)
- Construction schedule
- Construction budget

It can be very advantageous to involve a builder into the early stages of an HVAC design to bring their prospective and experience to the project system analysis and selection process. Whether a construction manager, design-builder, or integrated project delivery leader, each brings years of experience to the topic of constructability that can influence the final selection of HVAC system(s).

Construction budget constraints can also influence the choice of HVAC systems. Based on historical data, some systems may not be economically viable within the budget limitations of an owner's building program. In addition, annual maintenance and operating budget (utilities, labor, and materials) can be limiting although more often than not, annual operating budgets do not get included in the total construction "soft costs" (e.g., furnishing, general conditions). This can be particularly important for building owners who will retain the building for a substantial number of years. The building's estimator can assist in the overall project cost by introducing value-engineered solutions that may involve (1) cost-driven performance, which may provide a better solution for lower first cost; (2) a more sustainable solution over the life of the equipment; or (3) best value based on a reasonable return on investment.

Narrowing the Choices

The following chapters in this volume present information to help the design engineer narrow the choices of HVAC systems:

- **Chapter 2** focuses on a distributed approach to HVAC.
- **Chapter 3** provides guidance for large equipment centrally located in or adjacent to a building.
- **Chapters 4 to 18** address the numerous types of air-conditioning and heating systems.

Each chapter summarizes positive and negative features of various systems. Comparing the criteria, other factors and constraints, and their relative importance usually identifies one or two systems that best satisfy project goals. In making choices, notes should be kept on all systems considered and the reasons for eliminating those that are unacceptable.

Each selection may require combining a primary system with a secondary (or distribution) system. The primary system converts energy derived from fuel or electricity to produce a heating and/or cooling medium. The secondary system delivers heating, ventilation, cooling, humidification, and/or dehumidification to the occupied space. The systems are independent to a great extent, so several secondary systems may work with a particular primary system. In some cases, however, only one secondary system may be suitable for a particular primary system.

Once subjective analysis has identified one or more HVAC systems (sometimes only one choice remains), detailed quantitative evaluations must be made. All systems considered should provide satisfactory performance to meet the owner's essential goals. The design engineer should provide the owner with specific data on each system to make an informed choice. Consult the following chapters to help narrow the choices:

- Chapter 10 of the 2017 *ASHRAE Handbook—Fundamentals* covers physiological principles, comfort, and health.
- Chapter 19 of the 2017 *ASHRAE Handbook—Fundamentals* covers methods for estimating annual energy costs.
- Chapters 37 to 44 of the 2019 *ASHRAE Handbook—HVAC Applications* address important topics pertinent to the long-term success of building programs.

Other documents and guidelines that should be consulted are ASHRAE standards and guidelines; local, state, and federal guidelines; and special agency requirements (e.g., U.S. General Services Administration [GSA], Food and Drug Administration [FDA], Joint Commission on Accreditation of Healthcare Organizations [JCAHO], Facility Guidelines Institute [FGI], Leadership in Energy and Environmental Design [LEED®]). Additional sources of detailed information are listed in the Bibliography and/or available in the ASHRAE Bookstore (www.ashrae.org/bookstore).

Table 1 Sample HVAC System Analysis and Selection Matrix (0 to 10 Score)

Goal: Furnish and install an HVAC system that provides moderate space temperature control with minimum humidity control at an operating budget of 70,000 Btu/h per square foot per year				
Categories	System #1	System #2	System #3	Remarks
1. Criteria for selection: 78°F space temperature with $\pm 3^\circ\text{F}$ control during occupied cycle, with 40% rh and $\pm 5\%$ rh control during cooling. 68°F space temperature with $\pm 2^\circ\text{F}$, with 20% rh and $\pm 5\%$ rh control during heating season. - First cost - Equipment life cycle				
2. Important factors: - First-class office space stature - Individual tenant utility metering				
3. Other goals: - Engineered smoke control system - ASHRAE <i>Standard</i> 62.1 ventilation rates - Direct digital control building automation				
4. System constraints: - No equipment on first floor - No equipment on ground adjacent to building				
5. Energy use as predicted by use of an industry-acceptable computerized energy model				
6. Other constraints: No perimeter finned-tube radiation or other type of in-room equipment				
TOTAL SCORE				

Selection Report

As the last step, the design engineer should prepare a system analysis and selection report in a format that incorporates the following:

- Introduction
- Building program primary HVAC goals
- Criteria for selection
- Important factors, including advantages and disadvantages
- System integration with other building systems
- Other HVAC goals
- Security and environmental health and safety criteria
- Building envelope
- Project timeline (design, equipment delivery, commissioning, training, and construction)
- Basis of design
- HVAC system analysis and selection matrix
- System narratives
- Budget costs (first cost, operating cost, and energy cost)
- Final recommendation(s)

A brief outline of each of the final selections should be provided along with overall budget and/or cost per square foot. In addition, HVAC systems deemed inappropriate should be noted as having been considered but not found applicable to meet the owner's primary HVAC goal.

The report should include an HVAC system selection matrix that identifies the one or two suggested HVAC system selections (primary and secondary, when applicable), system constraints, and other constraints and considerations. In completing this matrix assessment, the design engineer should identify the owner's input to the analysis. This input can also be applied as weighted multipliers, because not all criteria carry the same weighted value.

Many grading methods are available to complete an analytical matrix analysis. Probably the simplest is to rate each item excellent, very good, good, fair, or poor. A numerical rating system such

as 0 to 10, with 10 equal to excellent and 0 equal to poor or not applicable, can provide a quantitative result. The HVAC system with the highest numerical value then becomes the recommended system to accomplish the goal.

The system selection report should include a summary followed by a more detailed account of the HVAC system analysis and system selection. This summary should highlight key points and findings that led to the recommendation(s). The analysis should refer to the system selection matrix (such as in [Table 1](#)) and the reasons for scoring.

With each HVAC system considered, the design engineer should note the criteria associated with each selection. Issues such as close-tolerance temperature and humidity control may eliminate some HVAC systems from consideration. System constraints, noted with each analysis, may eliminate potential HVAC systems. Advantages and disadvantages of each system should be noted with the scoring from the HVAC system selection matrix. This process should reduce HVAC selection to one or two optimum choices for presentation to the owner. Examples of similar installations for other owners can be included with this report to support the final recommendation. Identifying a third party for an endorsement allows the owner to inquire about the success of other HVAC installations.

2. HVAC SYSTEMS AND EQUIPMENT

Many built, expanded, and/or renovated buildings may be ideally suited for decentralized HVAC systems, with equipment located in, throughout, adjacent to, or on top of the building. The alternative to this decentralized approach is to use primary equipment located in a central plant (either inside or outside the building) with water and/or air required for HVAC needs distributed from this plant.

Decentralized System Characteristics

The various types of decentralized systems are described in [Chapter 2](#). The common element is that the required cooling is

distributed throughout the building, generally with direct-expansion cooling, electric, or gas-fired heating of air systems, and on occasion, oil-fired warm-air heating units.

Temperature, Humidity, and Space Pressure Requirements. A decentralized system may be able to fulfill any or all of these design parameters, but typically not as efficiently or as accurately as a central system.

Capacity Requirements. A decentralized system usually requires each piece of equipment to be sized for zone peak capacity, unless the systems are variable-volume. Depending on equipment type and location, decentralized systems do not benefit as much from equipment sizing diversity as centralized systems do.

Redundancy. A decentralized system may not have the benefit of back-up or standby equipment. This limitation may need review.

Facility Management. A decentralized system can allow the building manager to maximize performance using good business/facility management techniques in operating and maintaining the HVAC equipment and systems. With some decentralized equipment (e.g., rooftop HVAC units), operation and maintenance may be performed annually via a service contract with a mechanical contracting company.

Spatial Requirements. A decentralized system may or may not require equipment rooms. Because of space restrictions imposed on the design engineer or architect, equipment may be located on the roof and/or the ground adjacent to the building. Depending on system components, additional space may be required in the building for chillers and boilers. Likewise, a decentralized system may or may not require duct and pipe shafts throughout the building.

First Cost. A decentralized system probably has the best first-cost benefit. This feature can be enhanced by phasing in the purchase of decentralized equipment as needed (i.e., buying equipment as the building is being leased/occupied).

Operating Cost. A decentralized system can save operating cost by strategically starting and stopping multiple pieces of equipment. When comparing energy consumption based on peak energy draw, decentralized equipment may not be as attractive as larger, more energy-efficient centralized equipment.

Maintenance Cost. A decentralized system can save maintenance cost when equipment is conveniently located and equipment size and associated components (e.g., filters) are standardized. When equipment is located outdoors, maintenance may be difficult during bad weather.

Reliability. A decentralized system usually has reliable equipment, although the estimated equipment service life may be less than that of centralized equipment. Decentralized system equipment may, however, require maintenance in the occupied space.

Flexibility. A decentralized system may be very flexible because it may be placed in numerous locations.

Level of Control. Decentralized systems often use direct refrigerant expansion (DX) for cooling, and on/off or staged heat. This step control results in greater variation in space temperature and humidity, where close control is not desired or necessary. As a caution, oversizing DX or stepped cooling can allow high indoor humidity levels and mold or mildew problems.

Noise and Vibration. Decentralized systems often locate noisy machinery close to building occupants.

Constructability. Decentralized systems frequently consist of multiple and similar-in-size equipment that makes standardization a construction feature, as well as purchasing units in large quantities.

Centralized System Characteristics

These systems are characterized by central refrigeration equipment/systems with chilled-water distribution and central boiler equipment/system with hot-water distribution. Other central systems include low-, medium-, and high-pressure steam plants and

cogeneration central equipment/systems, or condenser water systems used with water-source heat pumps.

Among the largest centralized systems are HVAC plants serving groups of large buildings. These plants improve diversity and generally operate more efficiently, and with lower maintenance costs, than individual central plants. Economic considerations of larger centralized systems require extensive analysis. The utility analysis may consider multiple fuels and may also include gas and steam turbine-driven equipment. Multiple types of primary equipment using multiple fuels and types of HVAC-generating equipment (e.g., centrifugal and absorption chillers) may be combined in one plant.

Temperature, Humidity, and Space Pressure Requirements.

A central system may be able to fulfill any or all of these design parameters, typically with greater precision and efficiency than a decentralized system.

Capacity Requirements. A central system usually allows the design engineer to consider HVAC diversity factors that significantly reduce installed equipment capacity. As a result, this offers some attractive first-cost and operating-cost benefits.

Redundancy. A central system can accommodate standby equipment that decentralized configurations may have trouble accommodating because of cost.

Facility Management. A central system usually allows the building manager to maximize performance using good business/facility management techniques in operating and maintaining the HVAC equipment and systems. Operation and maintenance are usually handled by on-site staff, whether employees of the building owner or outsourced building management placed on site.

Spatial Requirements. The equipment room for a central system is normally located outside the conditioned area: in a basement, penthouse, service area, or adjacent to or remote from the building. A disadvantage of this approach may be the additional cost to furnish and install secondary equipment for the air and/or water distribution. Other considerations include access requirements and physical constraints that exist throughout the building to the installation of the secondary distribution network of ducts and/or pipes and for equipment replacement.

First Cost. Even with HVAC diversity, a central system may not be less costly than decentralized HVAC systems. Historically, central system equipment has a longer equipment service life to compensate for this shortcoming. Thus, a life-cycle cost analysis is very important when evaluating central versus decentralized systems.

Operating Cost. A central system usually has the advantage of larger, more energy-efficient primary equipment compared to decentralized system equipment. In addition, the availability of multiple pieces of HVAC equipment allows staging of this equipment operation to match building loads while maximizing operational efficiency.

Maintenance Cost. The equipment room for a central system provides the benefit of being able to maintain HVAC equipment away from occupants in an appropriate service work environment. Access to occupant workspace is not required, thus eliminating disruption to the space environment, product, or process. Because of the typically larger capacity of central equipment, there are usually fewer pieces of HVAC equipment to service.

Reliability. Centralized system equipment generally has a longer service life.

Flexibility. Flexibility can be a benefit when selecting equipment that provides an alternative or back-up source of HVAC.

Air Distribution Systems

The various air distribution systems, including dedicated outdoor air systems (DOAS), are detailed in [Chapter 4](#). Any of the preceding system types discussed within this chapter can be used in conjunction with DOAS.